Incident Response Investigation Using Wireshark

Introduction

In this project, I analyzed a packet capture (PCAPNG) file using Wireshark to understand the network activity occurring within the capture. The goal was to identify the protocols being used, examine the communicating endpoints, and determine whether any suspicious activity was present.

Tools Used

- Wireshark
- GitHub

Analysis Performed

1. Protocol Hierarchy Analysis

The Protocol Hierarchy statistics showed a mixture of network protocols, including:

- USB
- Ethernet
- IPv4 and IPv6
- TCP
- UDP
- DNS
- SMB
- Multicast DNS (mDNS)

Most of the traffic consisted of normal communication between devices and services on the system.

2. Endpoint Analysis

Using the Endpoints feature in Wireshark, I identified the most active IP addresses in the capture.

The most frequently observed addresses were:

- 127.0.0.1 (localhost)
- 127.0.1.1
- 192.168.2.1
- 192.168.2.255
- 224.0.0.22
- 224.0.0.251

These addresses are commonly associated with local system communication and local network activity.

3. DNS Traffic Analysis

I filtered the traffic using the `dns` filter to examine DNS requests and responses.

The capture showed DNS lookups for:

- ntp.ubuntu.com
- local

These requests appear to be related to normal hostname resolution and time synchronization services.

Findings

After reviewing the available traffic, I did not observe any obvious indicators of malicious activity.

There was no evidence of:

- Port scanning
- Malware communication
- Suspicious external connections
- Data exfiltration
- Unusual DNS activity

The captured traffic appeared to represent normal system and local network operations.

Conclusion

This investigation provided hands-on experience with Wireshark and basic incident response techniques. By examining protocol statistics, endpoint information, and DNS traffic, I was able to gain insight into how network communications can be analyzed.

Based on the evidence collected, the network activity in this capture appeared legitimate, and no suspicious behavior was identified during the investigation.